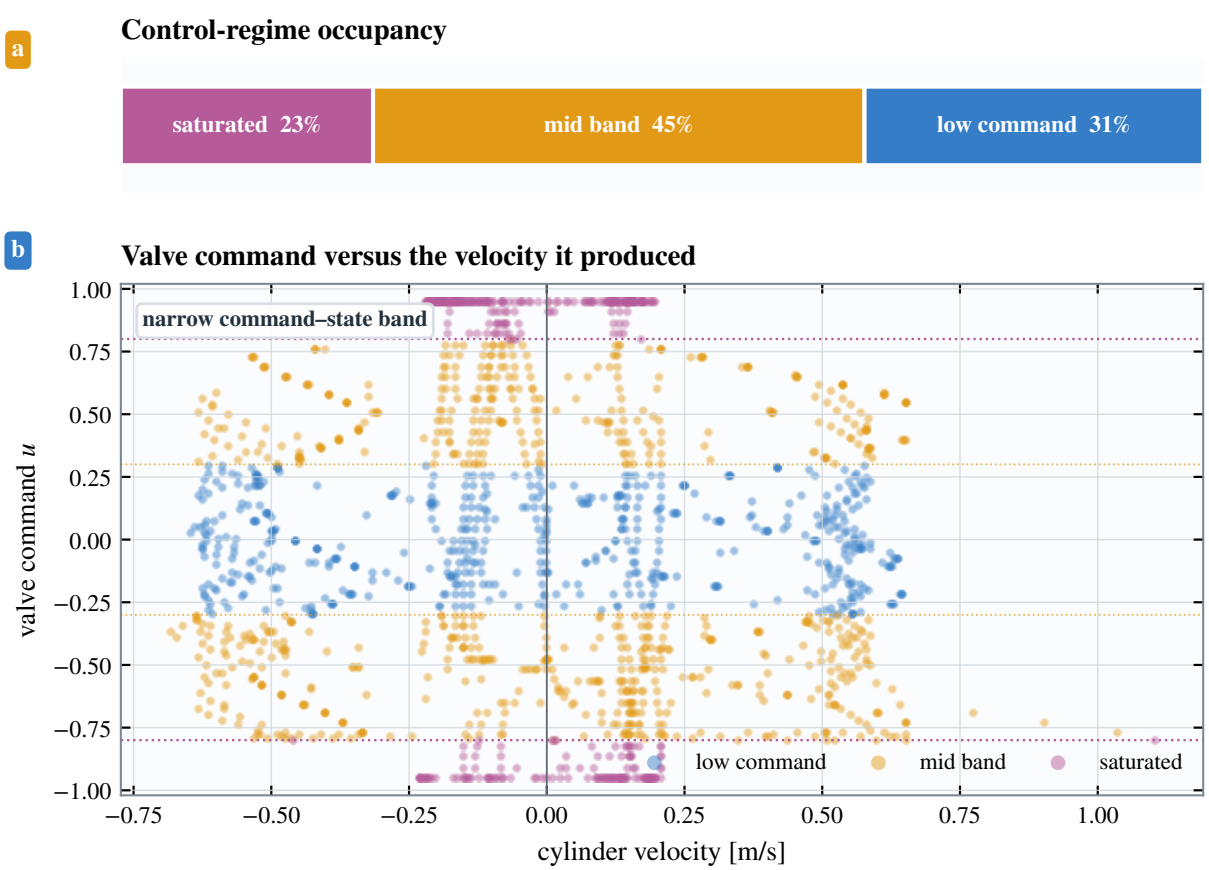


Figure 4.3: The closed-loop batch is diverse in control effort but not identifiable. (a) Fraction of time the valve spends saturated, in the mid band and in low-command creep. (b) The same run in the command–state plane, where the correlation that blocks identification is visible.

4.4 Open-Loop Excitation Experiments

4.4.1 Limitations of the Closed-Loop Batch

The first forward model achieved one-step $R^2 \approx 0.999$ on closed-loop data but was almost insensitive to valve input: zeroing the input changed long-horizon error by only a few percent (Section 5.3.1). In closed loop, command is largely determined by the state history already given to the model, so the data do not separate command causality from state correlation [Ljung 99].

Figure 4.3 confirms that coverage alone is not the issue. The valve is saturated for about one quarter of a pooled run, operates at mid-authority for roughly half, and otherwise remains in low-command creep. Despite this spread, command and cylinder velocity remain concentrated along a narrow band. The data are diverse in control effort but not identifiable, motivating a separate open-loop experiment.

4.4.2 Design of the Excitation Signal

The excitation run lasts 64 s and follows the fixed schedule in Figure 4.2(c). One second of startup is simulated and then discarded, because the plant is settling from its initial condition and those samples describe the initialisation rather than the dynamics. Four